TRI NHU DO

Curriculum Vitae

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Academic Experience

Research Experience

Mar. 2019 to Sep. 2019 **Postdoctoral Research Associate**, *The Univeristy of Texas at Dallas*, TX, USA.
Department of Computer Science
Our research focused on the theoretical performance of multiple-input multiple-output (MIMO) systems, in particular, the re-configurable distributed MIMO systems.
Supervisor: Prof. Zygmunt J. Haas (link)

Mar. 2013 to Aug. 2018 **Graduate Research Assistant**, *Hongik University*, South Korea.
Department of Software and Communications Engineering
Ubiquitous Networks & Convergence Laboratory (UNC Lab)

- I involved in writing research proposals, which resulted in the following grant:
  "*Development of energy efficiency and robust security-based cooperative relay transmissions and routing protocols for future mobile wireless networks,*" has been funded by the National Research Foundation of Korea (NRF), Grant No. 2016R1D1A1B03934898.
- I have been mentoring lab members of UNC Lab from Mar. 2016 to present.

Supervisor: Prof. Beongku An (link) and Prof. Daniel B. da Costa (link)

Teaching Experience

Mar. 2017 to Aug. 2018 **Teaching Associate**, *Hongik University*, South Korea.
Department of Software and Communications Engineering
Teaching Certificate (link)

Teaching undergraduate courses:

[012301]	Computer Engineering: Introduction and Practice	*Sping–2017/2018*
[704410]	Computer Architecture	*Fall–2017*

Education

Mar. 2015 to Aug. 2018 **Ph.D.**, *Hongik University*, South Korea.
Electronics and Computer Engineering

Mar. 2013 to Feb. 2015 **M.S.**, *Hongik University*, South Korea.
Electronics and Computer Engineering

Sep. 2007 to Feb. 2012 **B.Eng.**, *Posts and Telecommunications Institute of Technology*, Vietnam.
Electronics and Telecommunication Engineering

Publications

2019

- **Tri Nhu Do** and Zygmunt J. Haas, "On the Design of RD-MIMO: Spatial Multiplexing versus Opportunistic Transmission Schemes," (under preparing)

- Toan Van Nguyen, **Tri-Nhu Do**, Daniel Benevides da Costa, Vo Nguyen Quoc Bao, and Beongku An, "On the Performance of Multihop Cognitive Wireless Powered D2D Communications in WSNs," *IEEE Transactions on Vehicular Technology* (2nd round review with major revisions).

- **Tri-Nhu Do**, Van-Dinh Nguyen, Oh-Soon Shin and Beongku An, "Simultaneous Uplink and Downlink Transmissions for Wireless Powered Communication Networks," *IEEE Communications Letters*, vol. 23, no. 2, pp. 374-377, Feb. 2019.

- Toan-Van Nguyen, **Tri-Nhu Do**, Vo Nguyen Quoc Bao, Daniel Benevides da Costa, and Beongku An, "Performance Analysis of Multihop Cognitive WPCNs with Imperfect CSI," in *Proc. IEEE Global Communications Conference (GLOBECOM 2019)*, Waikoloa, Hawaii, USA, Dec. 2019.

- Toan-Van Nguyen, Van-Dinh Nguyen, **Tri-Nhu Do**, Daniel Benevides da Costa, and Beongku An, "Spectral Efficiency Maximization for Multiuser MISO-NOMA Downlink Systems with SWIPT," in *Proc. IEEE Global Communications Conference (GLOBECOM 2019)*, Waikoloa, Hawaii, USA, Dec. 2019.

2018

- **Tri-Nhu Do**, Daniel Benevides da Costa, Trung Q. Duong, and Beongku An, "Improving the Performance of Cell-Edge Users in NOMA Systems Using Cooperative Relaying," *IEEE Transactions on Communications*, vol. 66, no. 5, pp. 1883–1901, May 2018.

- **Tri-Nhu Do**, Daniel Benevides da Costa, Trung Q. Duong, and Beongku An, "Improving the performance of cell-edge users in MISO-NOMA systems using TAS and SWIPT-based cooperative transmissions," *IEEE Transactions on Green Communications and Networking*, vol. 2, no. 1, pp. 49–62, Mar. 2018.

- Nguyen Toan Van, **Tri-Nhu Do**, Vo Nguyen Quoc Bao, and Beongku An, "Performance analysis of wireless energy harvesting multihop cluster-based networks over Nakgami-m fading channels," *IEEE Access*, vol. 6, pp. 3068–3084, Feb. 2018.

- Kyusung Shim, **Tri-Nhu Do**, Beongku An, "Improving Physical Layer Security of NOMA Networks by Using Opportunistic Scheduling," in *Proc. 2018 International Conference on Ubiquitous and Future Networks (ICUFN)*, Prague, Czech Republic, Jul. 2018, pp. 262–267. —**Best Paper Award**

- Kyusung Shim, Hyukchun Oh, **Tri-Nhu Do**, Beongku An, "A Physical Layer Security-Based Transmit Antenna Selection Scheme for NOMA Systems," in *Proc. 2018 International Conference on Ubiquitous and Future Networks (ICUFN)*, Prague, Czech Republic, Jul. 2018, pp. 597–602.

- **Tri-Nhu Do**, and Beongku An, "Optimal sum-throughput analysis for downlink cooperative SWIPT NOMA systems," in *Proc. 2018 IEEE International Conference on Recent Advances in Signal Processing, Telecommunications & Computing (SigTelCom)*, Ho Chi Minh city, Vietnam, Jan. 2018, pp. 85–90.

- Kyusung Shim, **Tri-Nhu Do**, Beongku An, and Sang-Yep Nam, "A Wireless Power Transfer-based Routing Protocol for Heterogeneous Wireless Networks," in *Proc. 2018 International Conference on Green and Human Information Technology (ICGHIT)*, Chiang Mai, Thailand, Feb. 2018, pp. 75–79. —**Best Paper Award**

- Kyusung Shim, **Tri-Nhu Do**, and Beongku An, "A physical layer security-based routing protocol in mobile ad-hoc wireless networks," in *Proc. 20th IEEE International Conference on Advanced Communications Technology (ICACT)*, Gangwon-do, South Korea, Feb. 2018, pp. 417–422. —**Outstanding Paper Award**

2017

- **Tri-Nhu Do**, Daniel Benevides da Costa, Trung Q. Duong, Vo Nguyen Quoc Bao, and Beongku An, "Exploiting direct links in multiuser multirelay SWIPT cooperative networks with opportunistic scheduling," *IEEE Transactions on Wireless Communications*, vol. 16, no. 8, pp. 5410–5427, Aug. 2017.

- **Tri-Nhu Do**, Daniel Benevides da Costa, Trung Q. Duong, and Beongku An, "A BNBF user selection scheme for NOMA-based cooperative relaying systems with SWIPT," *IEEE Communications Letters*, vol. 21, no. 3, pp. 664–667, Mar. 2017.

- Kyusung Shim, **Tri-Nhu Do**, and Beongku An, "Performance analysis of physical layer security of opportunistic scheduling in multiuser multirelay cooperative networks," *Sensors*, vol. 17, no. 2, pp. 377–396, Feb. 2017.

- **Tri-Nhu Do** and Beongku An, "Finding optimal power splitting ratio for cooperative SWIPT NOMA systems," in *Proc. KSII The 9th International Conference on Internet (ICONI)*, Vientiane, Laos, Dec. 2017, pp. 135–136.

- **Tri-Nhu Do**, Daniel Benevides da Costa, Trung Q. Duong, and Beongku An, "A full-duplex cooperative scheme with distributed switch-and-stay combining for NOMA networks," in *Proc. 2017 IEEE Global Communications Conference (GLOBECOM)*, Singapore, Dec. 2017, pp. 1–6.

- **Tri-Nhu Do**, Daniel Benevides da Costa, Trung Q. Duong, and Beongku An, "Transmit antenna selection schemes for MISO-NOMA cooperative downlink transmissions with hybrid SWIPT protocol," in *Proc. 2017 IEEE International Conference on Communications (ICC)*, Paris, France, May 2017, pp. 1–6.

- Kyusung Shim, **Tri-Nhu Do**, and Beongku An, "Secure transmission under considering physical layer secure channel state information in mobile ad-hoc wireless," in *Proc. 2017 The Institute of Electronics Engineers of Korea (IEEK) Summer Conference*, Busan, South Korea, Jun. 2017, pp. 1282–1283.

- **Tri-Nhu Do**, Daniel Benevides da Costa, Trung Q. Duong, Vo Nguyen Quoc Bao, and Beongku An, "Opportunistic scheduling for fixed-gain amplify-and-forward-based multiuser multirelay SWIPT cooperative networks," in *Proc. 2017 IEEE International Conference on Recent Advances in Signal Processing, Telecommunications & Computing (SigTelCom)*, Da Nang, Jan. 2017, pp. 49–54.

2016

- **Tri-Nhu Do**, Daniel Benevides da Costa, and Beongku An, "Performance analysis of multirelay RF energy harvesting cooperative networks with hardware impairments," *IET Communications*, vol. 10, no. 18, pp. 2551–2558, Jun. 2016.

- Kyusung Shim, **Tri-Nhu Do**, and Beongku An, "The impact of hardware impairments and imperfect channel state information on physical layer security," *Journal of the Institute of Electronics and Information Engineers*, vol. 53, no. 4, p. 79–86, Apr. 2016.

- Kyusung Shim, **Tri-Nhu Do**, and Beongku An, "An analysis modeling architecture for supporting physical layer security of wireless networks under hardware impairments," *International Journal of Security and Its Applications*, vol. 10, no. 6, p. 351–362, 2016.

- **Tri-Nhu Do**, Vo Nguyen Quoc Bao, and Beongku An, "Outage performance analysis of relay selection schemes in wireless energy harvesting cooperative networks over non-identical rayleigh fading channels," *Sensors*, vol. 16, no. 3, pp. 295–314, Feb. 2016.

- Kyusung Shim, **Tri-Nhu Do**, and Beongku An, "A source relay selection scheme for physical layer security in multi-user multi-relay amplify-and-forward cooperative networks," in *Proc. 2017 International Conference on Green and Human Information Technology (ICGHIT)*, Hangzhou, China, Feb. 2016, pp. 339–340.

- Kyusung Shim, **Tri-Nhu Do**, and Beongku An, "Physical layer security under hard impairments in cooperative communication environments," in *Proc. 2016 The Institute of Electronics Engineers of Korea (IEEK) Summer Conference*, Jeju Island, South Korea, Jun. 2016, pp. 1641–1643.

- Kyusung Shim, **Tri-Nhu Do**, and Beongku An, "Optimal joint source and relay selection scheme for physical layer security in multi-user and multi-relay networks," in *Proc. KSII The 8th International Conference on Internet (ICONI)*, Hong Kong, Jul. 2016, pp. 187–188.

- Le The Dung, **Tri-Nhu Do**, and Beongku An, "Cooperative spectrum sensing based routing protocol in mobile cognitive radio ad-hoc networks," in *Proc. 2016 International Conference on Green and Human Information Technology (ICGHIT)*, Angeles City, Philippines, Feb. 2016, pp. 124–125.

- Kyusung Shim, **Tri-Nhu Do**, and Beongku An, "Physical layer security of wireless networks under hardware impairments," in *Proc. 2016 International Conference on Green and Human Information Technology (ICGHIT)*, Angeles City, Philippines, Feb. 2016, pp. 130–131.

- **Tri-Nhu Do**, Le The Dung, Beongku An, and Sang-Yep Nam, "Connectivity of hybrid overlay/underlay cognitive radio ad hoc networks," in *Proc. 2016 IEEE International Conference on Electronics, Information, and Communication (ICEIC)*, Da Nang, Vietnam, Jan. 2016, pp. 1–4.

- K. Shim, **Tri-Nhu Do**, Beongku An, and Sang-Yep Nam, "Outage performance of physical layer security for multi-hop underlay cognitive radio networks with imperfect channel state information," in *Proc. 2016 IEEE International Conference on Electronics, Information, and Communication (ICEIC)*, Da Nang, Vietnam, Jan. 2016, pp. 1–4.

2015

- **Tri-Nhu Do**, and Beongku An, "A soft-hard combination-based cooperative spectrum sensing scheme for cognitive radio networks," *Sensors*, vol. 15, no. 2, pp. 4388–4407, Feb. 2015.

- Kyusung Shim, **Tri-Nhu Do**, and Beongku An, "Physical layer security under hardware impairment and imperfect CSI in relaying networks," in *Proc. 2015 The Institute of Electronics Engineers of Korea (IEEK) Fall Conference*, Gwangju, South Korea, Nov. 2015, pp. 667-668.

- **Tri-Nhu Do** and Beongku An, "Outage analysis of best relay selection scheme for wireless energy harvesting relay networks," in *Proc. 2015 The Institute of Electronics Engineers of Korea (IEEK) Fall Conference*, Gwangju, South Korea, Nov. 2015, pp. 669–670.

- **Tri-Nhu Do**, Vo Nguyen Quoc Bao, and Beongku An, "A relay selection protocol for wireless energy harvesting relay networks," in *Proc. 2015 IEEE International Conference on Advanced Technologies for Communications*, Ho Chi Minh city, Vietnam, Oct. 2015, pp. 243–247.

- Kyusung Shim, **Tri-Nhu Do**, and Beongku An, "Secure transmission using DF on underlay cognitive radio," in *Proc. 2015 The Institute of Electronics Engineers of Korea (IEEK) Summer Conference*, Jeju Island, South Korea, Jun. 2015, pp. 1418–1420.

- **Tri-Nhu Do** and Beongku An, "A physical layer security-based multihop relaying protocol for underlay cognitive radio networks," in *Proc. 2015 The Institute of Electronics Engineers of Korea (IEEK) Summer Conference*, Jeju Island, South Korea, Jun. 2015, pp. 1454–1455.

- **Tri-Nhu Do** and Beongku An, "Secure transmission using decode-and-forward protocol for underlay cognitive radio networks," in *Proc. 2015 IEEE International Conference on Ubiquitous and Future Networks (ICUFN)*, Sapporo, Japan, Jul. 2015, pp. 914–918.

- **Tri-Nhu Do**, Beongku An, and Sang-Yep Nam, "A likelihood ratio test based hard combination scheme for cooperative spectrum sensing in cognitive radio networks," in *Proc. 2015 International Conference on Green and Human Information Technology (ICGHIT)*, Da Nang, Vietnam, Feb. 2015, pp. 195–196.

- **Tri-Nhu Do** and Beongku An, "A comparison of Neyman-Pearson and minimum sensing error criterions for cooperative spectrum sensing in cognitive radio networks," in *Proc. 2015 International Conference on Green and Human Information Technology (ICGHIT)*, Da Nang, Vietnam, Feb. 2015, pp. 181–182.

- **Tri-Nhu Do** and Beongku An, "Hybrid cooperative spectrum sensing scheme for cognitive radio networks," in *Proc. 2015 IEEE International Conference on Information Networking (ICOIN)*, Siem Reap, Cambodia, Jan. 2015, pp. 390–391.

2014

- **Tri-Nhu Do**, and Beongku An, "Cooperative spectrum sensing schemes with the interference constraint in cognitive radio networks," *Sensors*, vol. 14, no. 5, pp. 8037–8056, May 2014.

- **Tri-Nhu Do** and Beongku An, "A location information-based data fusion scheme for cooperative spectrum sensing in cognitive radio networks," in *Proc. 2014 The Institute of Electronics Engineers of Korea (IEEK) Fall Conference*, Seoul, South Korea, Nov. 2014, pp. 726–727.

- **Tri-Nhu Do** and Beongku An, "Log-likelihood ratio test based spectrum sensing for cognitive radio in small-scale fading channels," in *Proc. 2014 The Institute of Electronics Engineers of Korea (IEEK) Summer Conference*, Jeju Island, South Korea, Jun. 2014, pp. 1472–1473.

- **Tri-Nhu Do** and Beongku An, "Cooperative spectrum sensing schemes with the interference constraint in cognitive radio networks," in *Proc. 2014 IEEE International Symposium on Computer, Consumer and Control (IS3C)*, Taichung, Taiwan, Jun. 2014, pp. 1010–1013.

- **Tri-Nhu Do** and Beongku An, "Location-based user selection scheme for cooperative spectrum sensing in cognitive radio networks," in *Proc. 2014 International Conference on Green and Human Information Technology (ICGHIT)*, Ho Chi Minh City, Vietnam, Feb. 2014, pp. 355–356.

- **Tri-Nhu Do** and Beongku An, "Underlay approach-based soft combination schemes for cooperative spectrum sensing in cognitive radio networks," in *Proc. 2014 International Conference on Green and Human Information Technology (ICGHIT)*, Ho Chi Minh City, Vietnam, Feb. 2014, pp. 201–202.

- **Tri-Nhu Do**, Beongku An, and Kyu-Tae Lee, "A cooperative spectrum sensing scheme using the concept of underlay cognitive radio," in *Proc. 2014 IEEE International Conference on Electronics, Information, and Communication (ICEIC)*, Kota Kinabalu, Malaysia, Jan. 2014, pp. 1–3.

2013

- **Tri-Nhu Do** and Beongku An, "Cooperative spectrum sensing in cognitive radio networks under interference constraint," in *Proc. 2013 The Institute of Electronics Engineers of Korea (IEEK) Fall Conference*, Seoul, South Korea, Nov. 2013, pp. 902–903.

- **Tri-Nhu Do** and Beongku An, "A discussion of cooperative spectrum sensing using Alamouti code in cognitive radio networks," in *Proc. 2013 The Institute of Electronics Engineers of Korea (IEEK) Summer Conference*, Jeju Island, South Korea, Jun. 2013, pp. 1437–1438.

Honors and Awards

- **Top Reviewers Award** for Computer Science by Publons in Sep. 2018

- **Best Paper Award** in the 10th IEEE International Conference on Ubiquitous and Future Networks (ICUFN), Prague, Czech Republic, Jul. 2018

- **Best Paper Award** in the 6th International Conference on Green and Human Information Technology (ICGHIT), Chiang Mai, Thailand, Feb. 2018

- **Outstanding Paper Award** in the 20th IEEE International Conference on Advanced Communications Technology (ICACT), Gangwon-do, South Korea, Feb. 2018
- **Global Scholarship** awarded from Hongik University, South Korea Mar. 2015 – Dec. 2016
- **Global Scholarship** awarded from Hongik University, South Korea Mar. 2013 – Dec. 2014

Services

Technical Program Committee (TPC) Member

- 2019 IEEE Global Communications Conference
- 2018 IEEE 88th Vehicular Technology Conference: VTC2018-Fall
- 2018 IEEE International Conference on Advanced Networks and Telecommunications Systems

Reviewer

I have been serving as reviewer for top-tier journals and conferences as recored at *Review Record–Publons.com (link)*

References

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Prof. Beongku An

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Prof. Daniel B. da Costa

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